

UML DIAGRAM

DIAGRAM	PURPOSE
Use Case Diagram	Actors + ovals (use cases) connected by lines
Class Diagram	Rectangles divided into compartments: class name, attributes, methods
Activity Diagram	Flowchart-like: rounded actions, arrows, decision diamonds, start/end nodes
Sequence Diagram	Vertical lifelines + horizontal messages/arrows showing time/order
State Machine Diagram	States + transitions; rounded rectangles and arrows showing state changes
Component Diagram	Components/modules shown as rectangles with component symbol; interfaces may appear
Deployment Diagram	3D/node-looking boxes representing hardware/devices and deployed software
Object Diagram	Specific objects/instances, usually rectangles with underlined object names
Communication Diagram	Objects connected by links + numbered messages such as 1: login()
Package Diagram	Folder-shaped packages containing/grouping UML elements
Interaction Diagram	Shows interaction between objects; focuses on messages exchanged
Timing Diagram	Time is the main focus; horizontal timeline/waveform showing state/value changes
Composite Structure Diagram	Internal structure of a class/component, showing parts, ports, and connectors
Profile Diagram	Stereotypes, profiles, metaclasses, and extensions; often uses «stereotype» notation

- Actor + oval → USE CASE
- 3-part rectangle → CLASS
- Flowchart/diamond → ACTIVITY
- Lifelines + messages → SEQUENCE
- State changes → STATE MACHINE
- Software components → COMPONENT
- Physical nodes/devices → DEPLOYMENT
- Object instances + underlined names → OBJECT
- Numbered messages between objects → COMMUNICATION
- Folder shapes → PACKAGE
- Objects interacting/messages → INTERACTION
- Time/waveform → TIMING
- Inside structure/parts/ports → COMPOSITE STRUCTURE
- stereotypes/profile extensions → PROFILE

Use Case = Actors
Class = Blueprint
Activity = Workflow
Sequence = Lifelines
State = State changes
Component = Software pieces
Deployment = Hardware
Object = Instances
Communication = Numbered messages
Package = Folders
Interaction = Object communication
Timing = Time
Composite = Internal parts
Profile = Stereotypes

Sequence vs Communication

- **Sequence:** vertical lifelines and messages moving downward through time.
- **Communication:** objects are connected like a network, with numbered messages.

Class vs Object

- **Class:** general blueprint → `Student`, attributes, methods.
- **Object:** actual instance → `student1 : Student`, usually underlined.

Activity vs State Machine

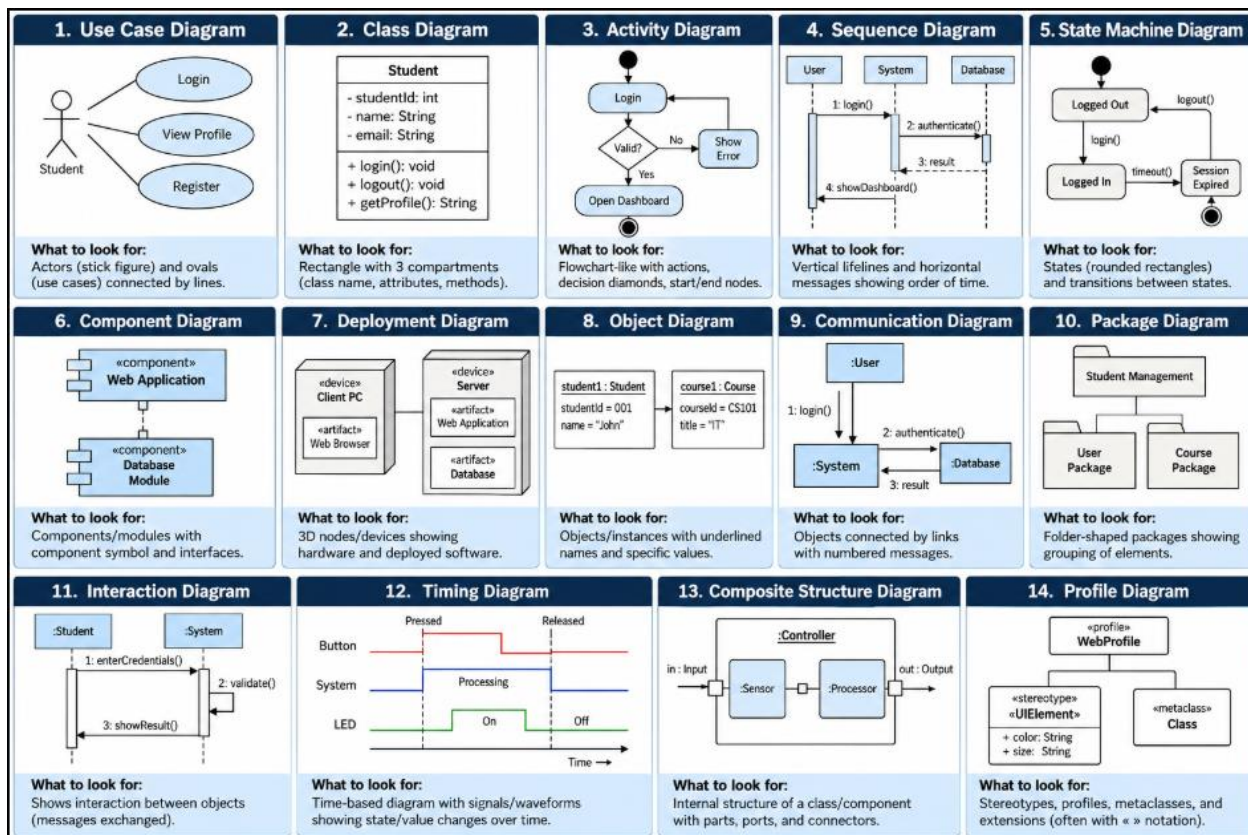
- **Activity:** represents a process/workflow.
- **State Machine:** represents states and how an object changes states.

Component vs Deployment

- **Component:** software/module structure.
- **Deployment:** physical hardware/nodes where software is deployed.

Interaction vs Sequence/Communication

- **Interaction** is the broader category concerned with objects communicating.
- **Sequence** and **Communication** are specific interaction diagram types.



- **Use Case** → 🧑 Actor + Ovals
- **Class** → □ 3-part rectangle
- **Activity** → ⚡ Flowchart + diamond
- **Sequence** → ↓ Vertical lifelines
- **State Machine** → ⚙ States + arrows
- **Component** → □ Software components
- **Deployment** → 🖨 3D devices/nodes
- **Object** → 📄 Underlined object names
- **Communication** → 📞 Objects + numbered messages
- **Package** → 📁 Folder shapes
- **Interaction** → 🗣 Objects exchanging messages
- **Timing** → 📊 Waveform + time
- **Composite Structure** → 🔧 Inside parts/ports
- **Profile** → «stereotype» notation